# Grove City College Status Sheet

Status Sheets are provided as a convenience for the student and may be helpful for recording completed courses. However, the College Bulletin is the controlling authority on all requirements. Questions should be directed to your academic advisor or the Registrar.

(WI)=Writing Intensive, (SI)=Speaking Intensive, (IL)=Information Literacy courses.

## B.S. in Physics/Computer with Software Option
## Entering in 2025

(REVISED 02-25-2025)

**Name:** _____

**ID#** _____ **Date:** _____

**Year of Anticipated Graduation:** _____ **Advisor:** _____

**TOTAL HOURS REQUIRED FOR THIS DEGREE---------- 128 HOURS**

**General Education + Elective Requirements----------38-40 HOURS**

**Minimum CQPA and MQPA required for graduation----------2.00**

**MQPA Courses----------ASTR; COMP; PHYS; ELEE; MATH 222**

**Major Requirements----------88-90 HOURS**

**GENERAL EDUCATION REQUIREMENTS---------- 26 HOURS**

| | | Cr. | Sem. Taken | Grade |
|---|---|---|---|---|
| **HUMANITIES CORE----------18 HOURS** | | | | |
| HUMA 100 | The Humanities: Christian Wisdom | 1 | | |
| HUMA 200 | Western Civilization | 3 | | |
| HUMA 202 | Civilization and Literature | 3 | | |
| HUMA 204 | Civilization and the Arts | 3 | | |
| HUMA 261 | Scripture & Theology for the Chr. Life I | 3 | | |
| HUMA 271 | Scripture & Theology for the Chr. Life II | 3 | | |
| HUMA 300 | Gospel & the Good Life: Christian Ethics | 2 | | |
| **WRITING REQUIREMENT----------3 HOURS** | | | | |
| WRIT 101 | Found. of Academic Discourse (WI/IL) | 3 | | |

**FOUNDATIONS OF THE SOCIAL SCIENCES*---------- 3 HOURS**

Choose one course from the following:

| | | |
|---|---|---|
| ECON 120 | Foundations of Economics | PSYC 101 Foundations of Psychology |
| HIST 120 | Foundations of History | PSYC 200 Cross-Cultural Psychology |
| HIST 204 | Historical & Phil. Found. of Education | SOCW 101 Foundations of Social Work |
| POLS 101 | Foundations of Political Science | SOCI 101 Foundations of Sociology |
| | | SOCI 103 Found. of Cultural Anthr. |

_____ 3 _____ _____

**NATURAL SCIENCE (with lab)/ QUANTITATIVE/LOGICAL REASONING---- 0 HOURS**

| | | |
|---|---|---|
| (1) Natural Science with lab | 4 | (Met through major) |
| (2) Quantitative/Logical Reasoning | 3-4 | (Met through major) |
| (3) Third course in Natural Science, Quantitative or Logical Reasoning | 3-4 | (Met through major) |

**STUDIES IN SCIENCE, FAITH, & TECHNOLOGY ---------- 2 HOURS**

Choose one course from the following:

| | |
|---|---|
| COMP 205/SSFT 205 | Ethics, Faith, and the Conscious Mind |
| PHIL 243 | Science and the Human: Inquiry, Design, & the Person |
| SSFT 210 | Science & Religion |
| SSFT 212 | Science, Faith, Technology, & Origins |

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**GENERAL ELECTIVES---------- 12-14 HOURS**

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**PHYSICS CORE REQUIREMENTS---------- 40 HOURS**

| | | Cr. | Sem. Taken | Grade |
|---|---|---|---|---|
| PHYS 101 | General Physics I - Engineering | 4 | | |
| PHYS 102 | General Physics II - Engineering | 4 | | |
| PHYS 135 | Horizons in Physics | 1 | | |
| PHYS 234 | Modern Physics | 3 | | |
| PHYS 242 | Introduction to Theoretical Physics | 3 | | |
| PHYS 288 | Intermediate Laboratory (WI) | 2 | | |
| PHYS 303 | Mechanics I | 3 | | |
| PHYS 321 | Radiation Laboratory (SI/IL) | 2 | | |
| PHYS 442 | Computational Methods in Physics | 3 | | |
| ASTR 207 | Introduction to Stars, Galaxies, & Cosmology | 3 | | |
| COMP 141 | Computer Programming I | 3 | | |
| COMP 244 | Database Management Systems | 3 | | |
| COMP 342 | Data Communication and Networking | 3 | | |
| COMP 424 | Parallel and Distributed Computing | 3 | | |

**TECHNICAL ELECTIVES---------- 3 HOURS**

Choose ONE of the following:

PHYS 304 Mechanics II, PHYS 305 Electricity and Magnetism; PHYS 340 Thermodynamics and Statistical Mechanics, PHYS 421 Advanced Topics, PHYS 431 Quantum Mechanics, ASTR 310 Astrophysics.

_____ 3 _____ _____

**COMPUTER SOFTWARE REQUIREMENTS----------25-27 HOURS**

| | | | | |
|---|---|---|---|---|
| COMP 220 | Computer Programming II | 3 | | |
| COMP 222 | Introduction to Data Structures/Algorithms | 3 | | |
| COMP 325 | Computer Architecture and Organization | 3 | | |
| COMP 340 | Operating Systems | 3 | | |
| COMP 350 | Software Engineering (WI/SI) | 3 | | |
| PHYS 210 | Electronics **OR** | | | |
| ELEE 201 Linear Circuits I & ELEE 251 Lab Skills & Prototyping | | 4 | | |

Choose TWO of the following (6-8 hours):

ELEE 204; COMP 480 or PHYS 270/370/470 (Limit 3 hrs); COMP 314 or any 400-level COMP course; MATH 213*, 222*.

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*If you complete MATH 213 and 222, you will receive a minor in Mathematics.

**TECHNICAL CORE REQUIREMENTS (Science, Math, etc.)---------- 20 HOURS**

| | | | | |
|---|---|---|---|---|
| CHEM 105 | Chemistry for Engineers | 4 | | |
| MATH 161 | Calculus I | 4 | | |
| MATH 162 | Calculus II | 4 | | |
| MATH 261 | Calculus III | 4 | | |
| MATH 262 | Differential Equations | 3 | | |
| MATH 263 | Numerical Differential Equations | 1 | | |

# SAMPLE FOUR-YEAR PLAN for the BACHELOR OF SCIENCE IN PHYSICS/COMPUTER with SOFTWARE OPTION

## Freshman Year

| Fall | Credits | Spring | Credits |
|---|---|---|---|
| PHYS 101 General Physics I | 4 | PHYS 102 General Physics II | 4 |
| PHYS 135 Horizons in Physics | 1 | COMP 220 Computer Programming II | 3 |
| COMP 141 Computer Programming I | 3 | MATH 162 Calculus II | 4 |
| MATH 161 Calculus I | 4 | Writing Requirement or HUMA Course | 3 |
| Writing Requirement or HUMA Course | 3 | Electives* | 2-3 |
| HUMA 100 The Humanities: Christian Wisdom | 1 | | 15 |
| | 16 | | |

## Sophomore Year

| Fall | Credits | Spring | Credits |
|---|---|---|---|
| CHEM 105 Chemistry for Engineers | 4 | PHYS 234 Modern Physics | 3 |
| MATH 213 Discrete Mathematics for Comp. Science*** | 3-4 | PHYS 242 Introduction to Theoretical Physics | 3 |
| MATH 261 Calculus III | 4 | PHYS 288 Intermediate Laboratory | 2 |
| HUMA Course | 3 | ASTR 207 Introduction to Stars, Galaxies, & Cosmology | 3 |
| SSFT Course | 2-3 | HUMA Course | 3 |
| | 17 | Electives* | 3 |
| | | | 17 |

## Junior Year

| Fall | Credits | Spring | Credits |
|---|---|---|---|
| PHYS 210 or ELEE 201 & 251 | 4 | PHYS 321 Radiation Laboratory | 2 |
| PHYS 303 Mechanics I | 3 | PHYS 442 Computational Methods in Physics** | 3 |
| COMP 222 Data Structures and Algorithms | 3 | COMP 342 Data Communication and Networking | 3 |
| Technical Elective | 3 | COMP 424 Parallel and Distributed Computing | 3 |
| HUMA Course | 3 | HUMA Course | 3 |
| | 16 | Electives* | 3 |
| | | | 17 |

## Senior Year

| Fall | Credits | Spring | Credits |
|---|---|---|---|
| COMP 244 Database Management Systems | 3 | COMP 340 Operating Systems | 3 |
| COMP 325 Computer Architecture and Organization | 3 | COMP 350 Software Engineering | 3 |
| Foundations of Social Science Course | 3 | MATH 262 Differential Equations | 3 |
| HUMA 300 Gospel and the Good Life: Christian Ethics | 2 | MATH 263 Numerical Differential Equations | 1 |
| Electives* | 4 | Electives* | 5 |
| | 15 | | 15 |

* 3-4 credits will be fulfilled by a Computer Software Requirement choice.

** The sequencing of MATH 262/263 and PHYS 442 may alternate because PHYS 442 is taught every other year.

*** Recommended Computer Software Requirement selection.